PROJECT REPOSITORY & DOCUMENTATION (README)

Business Operations & Performance Analysis — Portfolio Architecture

1. Project Title & Brief Overview

Project Title: Business Operations & Performance Analysis
Subtitle: Identifying Operational Inefficiencies, Measuring Business Performance, and Recommending Data-Driven Improvements
Author: Graduate in Information Science and Engineering (Fresher Operational Analyst)
Objective: This project serves as an end-to-end recruiter demonstration of operational data hygiene, multi-department KPI modeling, cost-efficiency diagnostics, and executive business storytelling for junior operations analyst roles.

2. Repository Structure & Artifact Catalog

Business_Operations_Performance_Analysis_Portfolio/
■■■ Business_Operations_Performance_Analysis.xlsx # Master 7-sheet formula workbook
■■■ Business_Operations_Project_Report.pdf # Full executive & technical report
■■■ Business_Operations_Clean_Data.csv # Single source of truth (240 clean rows)
■■■ Business_Operations_Monthly_Summary.csv # 12-month aggregated time series
■■■ Business_Operations_Department_Summary.csv # Departmental benchmark aggregates
■■■ Power_BI_Implementation_Guide.pdf # DAX and data modeling architecture
■■■ README_Business_Operations_Project.pdf # This comprehensive repository guide
■■■ Operational_Analyst_Project_Interview_Guide.pdf # 35+ interview Q&As;, elevator pitches & glossary

3. Tools, Methods & Technical Stack

Technology / Tool	Specific Techniques & Applications Used
Microsoft Excel	Advanced SUMIFS, COUNTIFS, AVERAGEIFS, dynamic charting, conditional alerting, multi-tab financial modeling.
Power BI & DAX	Dimensional modeling, Time-Intelligence DAX (CALCULATE, DATEADD, DIVIDE), KPI Card visuals, visual hierarchy.
Python (Pandas & OpenPyXL)	Synthetic data generation, reproducible statistical distributions, automated ETL cleaning log, programmatic workbook st
Business Operations Frameworks	Lean Six Sigma root-cause analysis, SLA compliance tracking, Cost-per-Unit unit economics, capacity utilization.

4. Key Analytical Insights Summary

• **95.90% Annual Achievement:** Nexora failed to meet scheduled target volumes across all 12 calendar months.
• **Service Delivery Drag:** Incurred 15.69% delay rate, ■385.45/unit cost, and generated 49.2% of all customer complaints.
• **Cost vs. Output Asymmetry:** Operating costs grew +18.89% while actual task completion rose only +9.81%, pointing to overtime inefficiency.
• **Quality Assurance Benchmark:** QA demonstrated highest operational discipline with 3.93% delay rate and lowest unit cost (■145.08).

5. Step-by-Step Reproduction Guide

1. **Excel:** Open Business_Operations_Performance_Analysis.xlsx in any modern spreadsheet application (Microsoft Excel 2016+, Office 365, or LibreOffice). Navigate to **START HERE**.
2. **Power BI:** Launch Power BI Desktop, import Business_Operations_Clean_Data.csv, and paste the measures from Power_BI_Implementation_Guide.pdf.
3. **Data Audit:** Review **DATA CLEANING** worksheet in Excel to inspect how nulls, duplicates, and casing errors were remediated.

6. Synthetic Data & Ethics Disclosure

This portfolio project was constructed using fully synthetic operational records generated to mirror real-world business challenges. All business names, figures, and departmental structures are fictional and designed strictly for skill demonstration.